

LServe + SampleAttention Slide Deck

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Post URL: [../blog/2026-01-30-lserve-sampleattention-sparse-attention/](https://blog/2026-01-30-lserve-sampleattention-sparse-attention/)

LONG-CONTEXT SERVING

Sparse Attention That Actually Survives Contact with Serving

A visual summary of what LServe and SampleAttention change in practice.

The shared thesis

**The win comes from structured sparsity
that kernels can exploit cheaply.**

LServe

Turns sparse attention into a serving-system design across prefill and decode.

Key idea

Use the right sparse structure for the actual bottleneck in each phase.

SampleAttention

Explains which sparse patterns matter during prefill and how to adapt them at runtime.

Key idea

Columns preserve global anchors.
Slashes preserve local continuity.

LServe Uses Different Sparsity for Prefill and Decode

One paper, two bottlenecks, two serving-time strategies.

Prefill

Goal: reduce sequential KV-block iterations.

How it works

1. Partition history into blocks.
2. Always keep the newest block.
3. Skip lower-value old blocks entirely.
4. Convert about half the heads into streaming heads.
5. Fuse sparse and standard heads in unified kernels.

Decode

Goal: reduce memory-bound KV-page reads for each query token.

How it works

1. Treat KV cache as pages.
2. Use a hierarchical selector to score which pages matter.
3. Reuse page-selection results across nearby decode steps.
4. Combine sparsity with KV quantization.

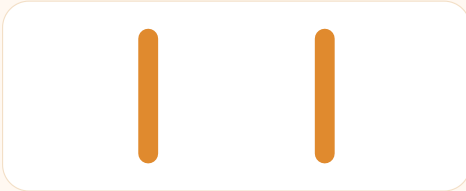
Mental model

Prefill skips blocks.
Decode prunes pages.

SampleAttention Says Two Sparse Patterns Matter Most

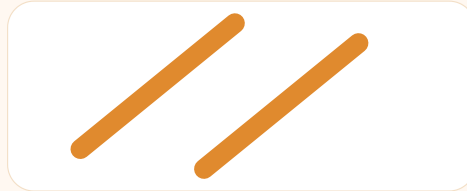
The paper's contribution is not just sparsity. It is runtime-adaptive sparsity with an accuracy

Column pattern



Semantic meaning:
Global anchors and attention sinks.
Many queries still need the same key positions.

Slash pattern



Semantic meaning:
Local continuity and recent context.
Queries attend along relative position bands.

CRA = Cumulative Residual Attention

Defined as the minimum remaining attention probability mass per query after sparsification.

Why it matters: CRA is a runtime proxy for whether the sparse mask kept enough useful attention or not. SampleAttention approximates it with query-guided chunked sampling, then picks top-k column and

What I Would Keep from Both Papers

A production-focused mental model for long-context inference.

1. Prefill and decode are different optimization problems.

If the bottlenecks differ, the sparse mechanism should differ too.

That is LServe's most useful systems lesson.

2. The best sparse patterns are structured, not arbitrary.

Columns preserve global anchors. Slashes preserve local continuity.

That is SampleAttention's cleanest conceptual contribution.

3. Accuracy needs a runtime control signal.

CRA is useful because it tracks how much important attention mass survives.

That is better than a fixed sparsity ratio or offline pattern choice.

Bottom line

SampleAttention tells you what to keep during prefill.

LServe tells you how to serve sparse attention across prefill and decode.